

VDA-MD FRAMEWORK

Value-Driven-AI Markdown Engine

Introducing C2MD: Compliance to Markdown

A new paradigm for AI governance: translating global compliance standards into human-readable, domain-owned, agent-actionable Markdown — governed by the customer journey that already exists in every organisation.

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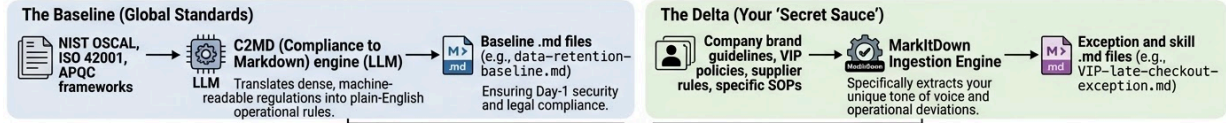
VDA-MD Framework

Architecture Overview

Value-Driven AI Markdown (VDA-MD) Framework

From Raw Compliance to Deterministic AI Execution and Auditability

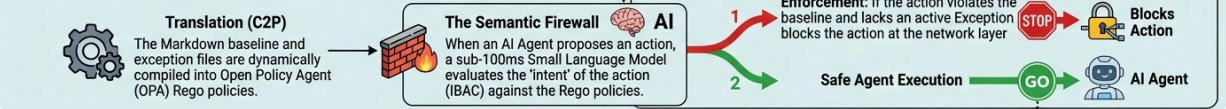
Layer 1: The Ingestion Layer: 'Baseline + Delta'



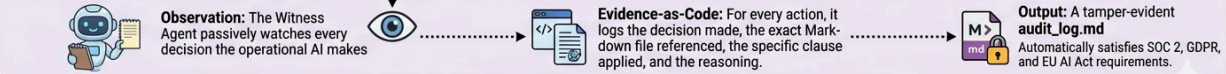
Layer 2: The Storage Layer: The Two-Axis Governance Map



Layer 3: The Execution Layer: A-Wrapper V2 Integration (Enterprise)



Layer 4: The Audit Layer: The Witness Agent



Executive Summary

The enterprise artificial intelligence landscape of 2026 is defined by a fundamental paradox: AI technology has never been more capable, yet more than 40% of enterprise agentic AI projects are projected to fail by 2027. The primary cause is not model quality. It is governance — specifically, the absence of a practical, scalable mechanism for encoding organisational rules, accountability, and compliance requirements into a form that AI agents can reliably act upon.

The VDA-MD Framework (Value-Driven-AI Markdown Engine) resolves this paradox through a novel architectural principle: instead of attempting to digitise decades of process debt from scratch, the framework establishes global best-practice standards as an automated baseline and uses AI to govern only what makes each organisation genuinely unique — its brand personality, operational exceptions, and domain-specific rules.

At the core of this framework is C2MD (Compliance to Markdown) — a new concept introduced in this document — which uses large language models to translate machine-readable compliance standards (NIST OSCAL, ISO 42001, APQC) into structured, human-readable Markdown files that serve as the operational governance layer for AI agents at runtime. This is distinct from and superior to existing Compliance-to-Policy (C2P) approaches, which generate machine-executable code that business owners cannot read, edit, or own.

The framework is delivered in two configurations. As a standalone product for SMBs and mid-market organisations, the VDA-MD requires no enterprise infrastructure — only a Claude API connection, a GitHub repository, and a simple dashboard. For enterprise deployments, the VDA-MD governance layer integrates directly with the A-Wrapper V2 execution engine (AgentGateway, Qdrant, OPA, ArgoCD) to produce the full A-VDA-Wrapper transformation system.

The organisational accountability architecture is built on the customer journey that already exists in every business. Vertical journey stages govern customer-facing operations; three universal horizontal value streams — Source-to-Pay, Order-to-Cash, and Onboarding-to-Final-Pay — govern the internal operational engine. Every AI agent in the organisation sits at the intersection of these axes, and that intersection determines which rules govern it, who owns those rules, and what approval chain applies to exceptions. The RACI is not a spreadsheet someone maintains — it is encoded in the file structure itself.

KEY FINDING	C2MD is an original concept with no prior art in published literature, open-source repositories, or standards body documentation as of March 2026. The term, the approach, and the specific combination of direction, ownership model, agent instruction purpose, and baseline-plus-delta architecture are first introduced in this document.
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1. The Problem: Process Debt and the AI Governance Gap

1.1 The Failure Rate Nobody Is Talking About

Despite significant executive enthusiasm and substantial investment, enterprise agentic AI adoption is encountering systemic failure. Market analysis indicates that over 40% of agentic AI projects are projected to be cancelled by 2027. Crucially, this failure is not attributable to model capability — the underlying foundation models are extraordinarily capable. The failure is architectural and organisational.

Organisations are attempting to deploy 2026-era autonomous AI systems atop governance frameworks designed for a world without AI agents. The result is a category of system that is technically sophisticated but organisationally ungovernable: agents that hallucinate because they lack authoritative context, agents that violate compliance because the rules are trapped in static documents no AI can read, and agents that behave inconsistently because no clear ownership model determines whose rules take precedence.

1.2 The Three Root Causes

Process Debt

Decades of operational rules — from VIP customer handling to supplier approval thresholds to employee onboarding procedures — are trapped in static, fragmented folders of PDFs, Word documents, and Excel sheets. These documents are not machine-readable. An AI agent cannot reliably govern its own behaviour against a 200-page PDF operations manual. Previous attempts to solve this by ingesting all process documentation have failed at cost and quality — the process debt is simply too large.

The Governance Technology Gap

Existing AI governance platforms (Credo AI, OneTrust, IBM Watsonx.governance, Collibra) are built for enterprise compliance officers with dedicated teams and large budgets. They produce dashboards that executives review, not systems that domain owners operate. They are passive monitoring tools, not active governance engines. None of them make the hotel general manager, the clinic owner, or the VP of Operations the actual operator of their AI governance.

The Compliance-to-Execution Disconnect

Compliance frameworks (NIST SP 800-53, ISO 42001, APQC process standards) exist in machine-readable formats (OSCAL XML, JSON). Policy enforcement engines (OPA, Kyverno) exist and are mature. But the bridge between compliance standards and runtime agent governance — the mechanism that translates 'what NIST requires' into 'what this agent must do before acting' — does not exist as a practical, deployable system. The Compliance-to-Policy (C2P) project generates Kubernetes infrastructure policies; it does not generate agent governance files that a business owner can read and a domain AI can act upon.

THE GAP

78% of organisations are using AI, but only 25-36% have a defined AI governance structure. The tools that exist are wrong for the problem: built for



compliance officers, not domain owners; producing reports, not governance files; monitoring humans, not governing agents.

2. The VDA-MD Framework: Core Architecture

2.1 The Founding Principle: Baseline Plus Delta

The VDA-MD Framework is built on a single architectural principle that thirty years of enterprise technology implementation has validated by observing the failures and embracing what works: assume best practice, govern only the exceptions, ensure ownership.

Previous enterprise transformation initiatives — Business Process Re-engineering in the 1990s, ERP implementations of the 2000s, Knowledge Management platforms of the 2010s — all failed at scale because they attempted to capture and digitise all organisational knowledge. The task is too large, too expensive, and too slow. By the time the digitisation is complete, the organisation has changed.

The VDA-MD inverts this model. It supplies a pre-configured, AI-generated baseline of global best practices anchored to NIST OSCAL, ISO 42001, and APQC process standards IN ENGLISH. This baseline covers the 80% of organisational behaviour that is structurally identical across all companies in the same industry. The Markdown ingestion engine and C2MD translation pipeline are then deployed exclusively to capture the 20% that makes each organisation genuinely unique: its brand voice, specific customer policies, supplier exceptions, and operational deviations.

The result is that an organisation can achieve AI governance and operational readiness in weeks, not years — because the vast majority of its governance baseline is pre-built, pre-validated, and immediately deployable in ENGLISH.

2.2 Markdown as the Governance Primitive

The strategic choice of Markdown as the universal governance format is not a technical convenience — it is the architectural insight that makes the entire system work. Markdown serves simultaneously as:

- A human-readable format that a VP of Operations, a Compliance Officer, or a hotel General Manager can read, understand, and edit without technical training
- The native instruction layer for large language models — empirical analysis demonstrates a 34-38% improvement in token efficiency over prose formats
- A versionable, diffable, committable artefact that Git treats as first-class text, providing the immutable audit trail that compliance frameworks require
- A structured format that preserves logical hierarchy through headings, YAML front matter, and section organisation while remaining close enough to plain text that no specialist tooling is required to view or edit it

This combination means that a single .md file simultaneously satisfies the needs of three different audiences: the business owner who authors it, the AI agent that governs itself against it at runtime, and the auditor who reviews it as compliance evidence. No other format achieves this.

2.3 The Four Core File Types

Every governance action in the VDA-MD system is governed by one of four Markdown file types, forming a complete separation of governance and execution:

File Type	Purpose	Example
AGENTS.md	Defines who the agents are, their scope of authority, and their inheritance hierarchy	Front-Desk-Bot.md — scope: Stay stage, inherits: Finance-Shared-Services
SOP.md	Human-readable policy — the rule that governs agent behaviour	VIP-late-checkout.md — VIP guests receive automatic 2pm late checkout subject to Finance approval
SKILL.md	Executable triggers — the API calls and system actions the agent may invoke	PMS-room-update.md — triggers Property Management System room status API
EXCEPTION.md	Approved deviations from the baseline, with owner, expiry, and approval metadata	EMEA-VIP-checkout.md — approved by COO, expires 31 Dec 2026, applies to Tier 1 EMEA accounts only

3. C2MD: Compliance to Markdown — A New Concept

3.1 What C2MD Is

Compliance to Markdown (C2MD) is an original concept introduced in this document. It defines the use of large language models to translate machine-readable compliance standards — specifically NIST OSCAL catalogs, ISO 42001 controls, and APQC process classification frameworks — into structured, human-readable, domain-owner-editable Markdown files that serve as the operational governance baseline for AI agents at runtime.

C2MD is explicitly distinguished from the existing Compliance-to-Policy (C2P) approach developed by IBM and the OSCAL Compass project. The following table defines the distinction precisely:

Dimension	C2P (Compliance to Policy)	C2MD (Compliance to Markdown)
Translation direction	OSCAL → Kyverno / Ansible / OPA Rego	OSCAL → structured human-readable .md
Output audience	Policy enforcement engines (machines)	Domain owners (humans) + AI agents
Output editability	Requires Rego/YAML expertise to modify	Any domain owner can read and edit
Runtime role	Infrastructure enforcement (Kubernetes clusters)	Agent governance baseline (LLM runtime)
Ownership model	Platform engineering team	VP of Operations, Compliance Officer, GM
Audit artefact	Machine-generated policy bundles	Human-approved, version-controlled .md files
Baseline coverage	IT infrastructure controls	All operational domains including S2P, O2C, HR
Maturity	Production for Kubernetes contexts (IBM/RedHat)	Original — first introduced March 2026

3.2 Why LLMs Are the Right Tool for C2MD

Large language models — specifically Claude — have deep, accurate training on NIST SP 800-53, ISO 42001, APQC process frameworks, and related compliance standards. This training means the model does not need to parse OSCAL XML and run it through a plugin architecture to understand what a control requires. It understands the control semantically, including its enhancement clauses, implementation guidance, and operational implications.

The C2P approach generates syntactically correct but semantically rigid policy code. The C2MD approach generates semantically rich, contextually aware Markdown that preserves the intent of

the control in language a domain owner can read and own. For the purpose of agent governance — where the agent must understand why a rule exists in order to apply it appropriately in edge cases — intent is what matters. A hotel General Manager does not need to know that NIST AC-2 maps to a Kyverno rule. She needs to know that all staff accounts must be reviewed quarterly and terminated within 24 hours of employee departure. The GM and the Agent are now working from the same English intent.

3.3 The C2MD Translation Pipeline

The C2MD pipeline operates in five stages:

1. Input ingestion: The NIST OSCAL JSON control (or ISO 42001 clause, or APQC process definition) is provided as structured context to the LLM, along with the vertical domain context (hospitality, logistics, financial services) and the specific operational area (Stay stage, Source-to-Pay, HR Onboarding).
2. Semantic translation: The LLM translates the control from regulatory language into operational language appropriate for the domain. The OSCAL control ID and framework reference are preserved in YAML front matter.
3. Agent rule extraction: The LLM extracts explicit agent decision rules from the translated control — what the agent must check before acting, what constitutes a violation, what requires human approval.
4. Confidence scoring: Each translated clause receives a confidence score. Clauses below threshold are flagged for human review, ensuring that complex multi-part controls are not silently oversimplified.
5. Human approval gate: The Compliance Officer reviews flagged clauses, corrects where necessary, and approves. The approval is logged in the YAML front matter with reviewer identity and timestamp. This approval record becomes the audit trail.

The output is a structured Markdown file with the following anatomy:

```
---
control_id: NIST-AC-2
framework: NIST SP 800-53 Rev 5
domain: HR / Onboarding-to-Final-Pay
journey_stage: Shared Services
owner: Head of HR
baseline: true
c2md_confidence: 0.94
reviewed_by: Jane Smith, Chief Compliance Officer
approved_date: 2026-03-18
expires: null
---

## Account management baseline

All employee system accounts must be formally requested by a line
manager before activation. Accounts must be deactivated within
```

24 hours of an employee's final working day. A quarterly review of all active accounts against current headcount is mandatory.

Agent rules

- Agent may NOT activate an account without a logged manager request recorded in the HR system
- Agent must trigger human review if deactivation is not confirmed within 48 hours of the recorded departure date
- Agent must escalate immediately for any active account belonging to a departed employee beyond 30 days post-departure

Violation definition

Any active account for a departed employee beyond 24 hours constitutes a control violation requiring Compliance Officer notification and immediate remediation action.

PRIOR ART NOTE

A search conducted on March 19, 2026 returned zero results for 'C2MD', 'Compliance to Markdown', or any equivalent named approach combining OSCAL/LLM translation with human-readable agent governance files. NIST raised a related concept in 2019 (Controls Markdown Language) as a structural reformatting request — never implemented and directionally different. IBM's 2025 OSCAL workshop noted LLMs could automate policy code generation — in the opposite direction (to machine-executable code, not human-readable governance). C2MD as defined in this document is original.

4. The VDA Customer Journey Model: Organisational Accountability Architecture

4.1 The Two-Axis Governance Map

The VDA customer journey model is not a UX framework or a marketing taxonomy. It is an organisational accountability architecture — the mechanism by which the RACI for AI governance is encoded into the file structure itself rather than maintained as a separate spreadsheet.

The framework establishes a two-axis governance map. The vertical axis represents customer-facing value streams — the journey stages that define how the organisation delivers its proposition to customers. The horizontal axis represents the universal operational engine — the three shared service value streams that sustain every business regardless of industry.

Every AI agent in the organisation sits at a specific intersection of these two axes. That intersection determines which Markdown files govern its behaviour, who owns those files, and what cross-domain approval chain applies when the agent needs to invoke capabilities that cross domain boundaries.

4.2 Vertical Journey Stages (Hospitality Example)

The vertical journey stages represent the customer-facing value stream. Each stage is owned by a named domain owner — a business person, not a technical person. The example below uses the hospitality vertical, but the pattern applies **universally** to any industry by substituting the appropriate stage labels.

Journey Stage	Domain Owner / Description
Pre-Book	Marketing Director — AI agents governing discovery, availability queries, pricing display, and initial engagement
Book	Revenue Manager — AI agents governing reservations, payment capture, confirmation workflows, and upsell offers
Stay	Operations Director — AI agents governing check-in, room management, guest requests, service delivery, and complaint handling
Post-Stay	CX Director — AI agents governing checkout, review solicitation, loyalty crediting, and re-engagement campaigns
Shared Services: Legal	General Counsel — cross-cutting legal constraints inherited by all vertical agents
Shared Services: Finance	CFO — financial authority limits and approval thresholds inherited across all stages
Shared Services: Tech	CTO — system access, integration permissions, and data handling rules

Shared Services: HR	CHRO — people-related governance inherited by all agents with people-facing actions
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4.3 The Cross-Domain Inheritance Rule

The critical architectural mechanism that distinguishes the VDA journey model from a simple file taxonomy is the cross-domain inheritance rule: a vertical agent cannot execute an action that crosses into a Shared Services domain without inheriting explicit permission from the relevant Shared Services Markdown files.

A Stay agent handling a room upgrade request cannot grant a financial concession without the Shared Services: Finance SOP.md permitting it. A Book agent capturing payment cannot store card data without the Shared Services: Legal SOP.md defining the permitted retention policy. A Post-Stay agent processing a refund cannot exceed the threshold defined in the Shared Services: Finance exception files without triggering a human approval workflow.

This is not a policy written in a separate governance document. It is structurally enforced by the agent's instruction files at runtime. The Witness Agent logs every cross-domain inheritance check, creating an automatic audit trail that demonstrates the governance structure was actually applied, not just documented.

RACI INSIGHT

The RACI emerges from the file structure automatically. Finance owns all S2P-* and O2C-* files. HR owns all HRpay-* files. Marketing owns all pre-book-* files. The intersection points — where a vertical agent invokes a shared service rule — are exactly where cross-functional approval is architecturally required. No RACI spreadsheet needs to be maintained separately.

5. The Three Universal Horizontal Value Streams

5.1 Why These Three

Source-to-Pay, Order-to-Cash, and Onboarding-to-Final-Pay were selected as the universal horizontal value streams because they share a property that makes the C2MD baseline model particularly powerful: they are structurally identical across all organisations in the same industry at the 80-85% level.

The process of matching a supplier invoice to a purchase order follows the same logic whether the organisation is a brewery or a hotel chain. The steps to onboard an employee are structurally identical whether the company is headquartered in Amsterdam or Auckland — with local statutory variation representing precisely the kind of exception delta that the VDA-MD ingestion engine is designed to capture. This structural universality means pre-built C2MD baselines for these three value streams can be genuinely comprehensive before a single customer document is ingested.

5.2 Source-to-Pay (S2P)

The Source-to-Pay value stream covers the complete procurement lifecycle from need identification through supplier selection, purchase order creation, goods receipt, invoice matching, and payment execution. The C2MD baseline for S2P is anchored to:

- NIST SP 800-53 SA-4 (Acquisition Process) for procurement governance
- APQC Process Classification Framework Category 4.0 (Manage Supply Chain)
- CIPS best practice standards for supplier qualification and approval
- Three-way match controls (PO, goods receipt, invoice) as universal baseline

The exception delta for S2P captures what makes each organisation unique: approval authority thresholds by spend category, preferred supplier lists that bypass standard sourcing requirements, country-specific payment terms, and emergency procurement procedures. A CFO who needs to change the approval threshold for a supplier category edits the relevant S2P-approval-authority.md file. The Sandbox runs the change against synthetic purchase orders before deployment. The audit trail is automatic.

5.3 Order-to-Cash (O2C)

The Order-to-Cash value stream covers the complete customer revenue lifecycle from quote generation through order management, credit checking, fulfilment, invoicing, accounts receivable, collections, and cash application. The C2MD baseline for O2C is anchored to:

- NIST SP 800-53 AU-2 (Event Logging) for transaction auditability
- APQC Category 2.0 (Develop and Manage Products and Services) for order management
- PCI DSS controls for payment data handling as a universal baseline
- Standard credit control procedures and collections escalation paths

The exception delta for O2C captures customer-specific pricing tiers, strategic account credit limit exceptions, territory-specific billing rules, and dispute resolution authority levels. Because the O2C baseline handles financial transactions, cross-domain inheritance from Shared Services: Finance is mandatory for any exception involving credit limits, payment terms, or refund authorisation above standard thresholds.

5.4 Onboarding to Final Pay (HRpay)

The Onboarding-to-Final-Pay value stream covers the complete employee lifecycle from offer acceptance through onboarding, role changes, performance management, benefits administration, payroll processing, and final settlement. This is the most heavily regulated of the three value streams, governed by employment law, GDPR, local labour codes, and sector-specific requirements. The C2MD baseline is anchored to:

- NIST SP 800-53 AC-2 (Account Management) for system access governance
- GDPR Article 5 data minimisation principles for employee data handling
- ISO 42001 human oversight requirements for AI systems making people-affecting decisions
- Standard onboarding and offboarding procedures as universal baseline

The exception delta for HRpay captures local statutory variations (jurisdiction-specific notice periods, local payroll cycle requirements), grade-specific benefit packages, company-specific probation policies, and collective agreement deviations. Because HRpay agents may take actions that affect employee rights, any exception to the baseline requires Shared Services: Legal inheritance and, for high-risk actions, a mandatory human-in-the-loop approval workflow.

5.5 File Naming Convention

The naming convention for all Markdown files encodes the value stream stage explicitly. This matters for auditability: when the Witness Agent writes its audit log, the file name alone should identify which stage of which value stream governed which agent action — without requiring the auditor to open the file or read its YAML metadata.

File Name Pattern	Example
S2P-[stage]-[rule].md	S2P-approval-authority.md, S2P-supplier-qualification.md, S2P-PO-threshold-EMEA.md
O2C-[stage]-[rule].md	O2C-credit-limit-standard.md, O2C-collections-escalation.md, O2C-strategic-account-exception.md
HRpay-[stage]-[rule].md	HRpay-onboarding-system-access.md, HRpay-offboarding-24hr-termination.md, HRpay-payroll-cycle-NL.md
[vertical]-[stage]-[rule].md	hospitality-stay-VIP-checkout.md, logistics-delivery-exception-handling.md
shared-[domain]-[rule].md	shared-finance-approval-matrix.md, shared-legal-data-retention.md

6. Domain Ownership, RACI, and Exception Management

6.1 The Machine-Enforced RACI

Traditional RACI matrices are governance theatre: consultants produce them, executives approve them, and then they sit in SharePoint while the actual decision-making follows informal patterns that nobody documents. The VDA-MD Framework replaces the RACI spreadsheet with a machine-enforced RACI encoded in the YAML front matter of every Markdown file.

The four RACI roles are represented as explicit metadata fields:

```
---
control_id: S2P-approval-authority
domain: Shared Services / Finance
owner: CFO                                # Accountable
authored_by: Head of Procurement         # Responsible
consulted: [Legal, CISO]                 # Consulted
informed: [COO, Compliance Officer]      # Informed
approved_by: CFO
approved_date: 2026-03-18
expires: 2027-03-18
risk_level: high
---
```

When an exception is proposed, the VDA workflow engine reads the YAML metadata of the relevant baseline file to determine the approval chain. A change to an S2P financial threshold automatically routes to the CFO for approval, notifies Legal and the CISO for consultation, and informs the COO and Compliance Officer upon deployment. No human needs to remember or enforce this routing — it is structurally determined by the file's metadata.

6.2 Exception Lifecycle Management

Every exception in the VDA-MD Framework is subject to a formal lifecycle with four mandatory stages:

6. Proposal: The domain owner proposes an exception via the VDA dashboard, specifying the baseline rule being deviated from, the conditions under which the exception applies, the business justification, the proposed expiry date, and the risk acceptance criteria.
7. Sandbox validation: The exception is tested against 50-100 synthetic agent interactions in the Sandbox before any human is asked to approve it. Conflicts with other rules (such as a late-checkout exception conflicting with a system IoT key expiry rule) are automatically surfaced at this stage.
8. Approval routing: Based on the YAML metadata of the affected baseline file, the system routes the exception to the appropriate approver(s), with consultation requests sent to the relevant stakeholders. All approvals are time-stamped and logged.

9. Expiry management: Every exception carries a mandatory expiry date. Thirty days before expiry, the system notifies the domain owner and the accountable executive. If the exception is not actively renewed, it expires automatically and the baseline rule is reinstated.

**COMPLIANCE
INSIGHT**

The exception lifecycle produces compliance evidence as a byproduct of operations. Every approved exception has a dated approval record, a risk acceptance statement, and a sandbox validation log. When an auditor requests proof that exceptions are properly controlled, the VDA dashboard produces the complete evidence trail automatically — not from a separate compliance system, but from the governance system that runs daily operations.

6.3 The Witness Agent

The Witness Agent runs in parallel with every operational AI agent in the VDA-MD system. It does not make decisions — it observes and records. For every agent action, the Witness Agent writes a structured entry to the domain's `audit_log.md` recording:

- Timestamp and agent identity
- The action proposed and the action taken
- Every Markdown file consulted in reaching the decision, with the specific clauses referenced
- The cross-domain inheritance checks performed and their outcomes
- Whether a human approval workflow was triggered and, if so, who responded and when
- The final decision rendered and the rule basis for that decision

This log is not generated for auditors — it is generated as a natural side-effect of how the agent operates. The auditor receives an automatic, complete, and tamper-evident record of exactly how every agent decision was made, against which version of which governance file, approved by whom, and on what basis. This is Evidence-as-Code: compliance proof generated continuously, not assembled retrospectively.

7. Product Architecture: Standalone and Enterprise

7.1 VDA-MD Standalone: For SMBs and Mid-Market

The VDA-MD standalone product is designed for organisations of 5 to 500 people who need AI governance & operations but do not have enterprise infrastructure teams, platform engineers, or dedicated compliance departments. The architecture is deliberately minimal — using only infrastructure that any organisation already has or can access as a managed service.

Component	Technology / Rationale
AI reasoning engine	Claude API (Anthropic) — handles C2MD translation, agent reasoning, gap analysis, and exception drafting. No local model required.
Document ingestion	Microsoft MarkItDown (MIT, v0.1.5) — converts PDFs, DOCX, XLSX, PPTX into clean Markdown. Deployed as an MCP server. Ingestion of Brand/Corporate Secret Sauce is the scope.
Governance ledger	GitHub private repository — immutable version history for all .md files satisfying SOC 2 audit trail requirements. Abstracted from end users. Can be used for immaculate version control and thereby change management.
Dashboard	React-based web application — non-technical interface replacing Git terminology with business language. 'Save Version' not 'Commit'.
Sandbox engine	Isolated simulation environment — 50-100 synthetic agent interactions against proposed governance changes before live deployment. “What If” impact modelling.
Standards baseline	GDPR, EU AI Act, NIST OSCAL + ISO 42001 + APQC — machine-readable compliance sources used by C2MD pipeline. Pre-loaded per vertical as ENGLISH.
Runtime	Claude API at inference time — agents read their governing .md files and reason over them directly. No OPA required for standalone. Exceptions to Human in the Loop (HITL).
Audit log	Witness Agent writing to audit_log.md — Evidence-as-Code generated as operational byproduct.

The key architectural insight for the standalone product is that Claude is simultaneously the C2MD translation engine, the Sandbox simulation engine, the gap analysis engine, and the runtime governance engine. The .md files it produces during setup become the instruction layer it reads during operation. This eliminates the need for a separate policy enforcement infrastructure (OPA, Rego, Kubernetes) that SMB organisations cannot practically operate.

7.2 A-VDA-Wrapper: The Enterprise Configuration

For enterprise deployments — organisations with dedicated platform engineering teams, existing Kubernetes infrastructure, and regulatory requirements for deterministic policy enforcement at the network layer — the VDA-MD governance layer integrates with the A-Wrapper V2 execution engine to produce the full A-VDA-Wrapper architecture.

In this configuration, the .md files produced by C2MD serve a second function beyond runtime agent instruction: they feed a second AI step that generates OPA Rego policies from the Markdown baseline and exception files. This produces deterministic, fail-closed enforcement at the network layer via the Semantic Firewall, while preserving the Markdown as the human-readable source of truth that business owners can read and edit without engineering intervention.

VDA-MD Component	A-Wrapper V2 Integration
C2MD baseline .md files	Feed Rego generation pipeline → OPA policies enforced at Semantic Firewall
Exception .md overlay	Compiled as Rego overlay on top of baseline via C2P plugin → zero-downtime deployment via Argo Rollouts
GitHub ledger	ArgoCD GitOps pipeline source of truth → automated deployment to AgentGateway
Witness Agent logs	OpenTelemetry integration → SIEM (Splunk/Datadog) for enterprise security operations
Sandbox simulation	Progressive delivery via Canary Rollouts — validated by Semantic Firewall intent telemetry
VDA dashboard	Non-technical interface to Qdrant vector state, Entra ID RACI, and ArgoCD deployment status

7.3 The Upgrade Path as Business Model

The most commercially significant architectural decision in the VDA-MD Framework is that the upgrade path from standalone to enterprise is non-destructive. An SMB that starts on the VDA-MD standalone product — building its .md governance library, training its domain owners to author and own their files, and operating its Witness Agent — does not discard that investment when it grows into enterprise-scale AI deployment.

The .md files port directly. The RACI metadata ports directly. The journey map and domain ownership model port directly. The organisation moves from Claude-as-runtime-engine to OPA-Rego-as-deterministic-enforcement without rewriting its governance logic — because the governance logic was always in the Markdown, and the Markdown is format-independent. This creates genuine switching cost not through lock-in, but through accumulated governance quality that the organisation has built and owns.

8. Proof of Concept Design

8.1 Scope and Objectives

The POC is designed to validate three specific hypotheses that are unproven at the time of this document's publication. Everything else in the framework relies on mature, production-grade components. These three hypotheses are the only genuine unknowns:

- C2MD quality hypothesis: An LLM (Claude) can translate NIST OSCAL controls into operational Markdown governance files with sufficient fidelity (>85% clause accuracy) that a domain expert review constitutes a practical quality gate.
- Exception overlay hypothesis: A Markdown-defined exception correctly overrides a Markdown baseline when both are provided as LLM context at agent decision time, without the agent conflating the two or applying the exception outside its specified conditions.
- Audit trail hypothesis: The Witness Agent produces an audit log that a Compliance Officer would accept as sufficient evidence of governance compliance without additional documentation.

The POC uses the hospitality vertical (Stay stage) for the vertical journey and Source-to-Pay for the horizontal value stream. This combination is chosen because the hospitality Stay stage has concrete, testable agent interactions (room upgrades, late checkouts, complaint handling) and S2P has clear numerical thresholds that make exception boundary testing straightforward.

8.2 Phase 1: Ingestion Quality

Five documents of mixed format are ingested through Markdown and processed by the C2MD tagging pipeline. The document mix deliberately includes formats that stress-test Markdown's limitations:

- Clean Word document SOP — front desk procedures. Expected: high fidelity.
- Excel sheet — supplier approval thresholds. Expected: high fidelity for tabular data.
- Scanned PDF — legacy room upgrade policy. Expected: OCR challenges; validates human review flag.
- PowerPoint deck — brand guidelines. Expected: moderate fidelity; tests image handling.
- Corporate exception document — existing approved policy deviations. Expected: high fidelity.

Success criterion: 4 of 5 documents produce .md files where a domain expert confirms the governance rule was captured correctly. The scanned PDF failure is expected and validates the human review flagging mechanism. The Excel sheet is the critical test — S2P approval thresholds live in spreadsheets.

8.3 Phase 2: C2MD Baseline Generation

Three NIST SP 800-53 controls are translated through the C2MD pipeline into operational baseline Markdown files:

- AC-2 (Account Management) — maps to HRpay Onboarding-to-Final-Pay domain
- AU-2 (Event Logging) — maps to Witness Agent audit log specification
- SA-4 (Acquisition Process) — maps to S2P purchase approval baseline

For each control, the C2MD pipeline generates a structured .md file with confidence scores per clause. The Compliance Officer reviews all flagged clauses and provides corrections. The corrected files are approved and committed with reviewer metadata.

Success criterion: A Compliance Officer who has never seen OSCAL can read each resulting .md file, confirm it accurately represents the NIST control in operational terms, and approve it for use as a governance baseline without reference to the NIST source document. If the source document is required for understanding, the translation has failed.

8.4 Phase 3: Exception Overlay and Boundary Testing

One baseline .md (S2P purchase approval authority from Phase 2) and one exception .md (EMEA VIP supplier pre-approved up to €75k without standard three-quote process) are provided as agent context. The agent is tested against five synthetic purchase order scenarios designed to probe boundary conditions:

Test Scenario	Expected Decision
Standard supplier, €10k purchase	PASS — baseline permits, no exception applies
Standard supplier, €80k purchase	FAIL — exceeds baseline threshold, CFO approval required
EMEA VIP supplier, €60k purchase	PASS — exception applies within threshold
EMEA VIP supplier, €80k purchase	FAIL — exceeds exception threshold, reverts to baseline
EMEA VIP supplier, €60k, request dated Jan 2027	FAIL — exception expired 31 Dec 2026

Success criterion: All five tests produce the correct agent decision with the Witness Agent log correctly identifying which file (baseline or exception) governed the decision and why. If 4/5 pass, the architecture is sound but prompt engineering requires refinement. If 3/5 or fewer pass, the LLM-as-runtime-engine hypothesis for exception boundaries requires fundamental rethinking.

8.5 Phase 4: Witness Agent and Audit Trail

The Witness Agent is wired to all test scenarios from Phase 3. Every decision generates a structured audit_log.md entry. The complete log from all five test scenarios is presented to a Compliance Officer with a single question: if an auditor asked you to prove that this supplier payment was properly authorised, could you answer using only this log?

Success criterion: Yes, without qualification. The log must identify the authorising rule, the file that contained it, who approved that file, when it was approved, whether an exception applied and under what conditions, and the final decision rendered.

8.6 What POC Success Proves

A successful POC across all four phases proves the single most important claim of the entire framework: that a large language model can serve reliably as both the C2MD translation engine (producing governance files from compliance standards) and the runtime governance engine (applying those files correctly at decision time), with a human review gate providing the quality assurance that prevents silent errors from propagating into production governance.

Everything else in the framework — the journey map, the RACI encoding, the Witness Agent, the exception lifecycle, the MarkItDown ingestion — is either already proven or has no fundamental uncertainty. The C2MD translation quality and the LLM exception boundary reliability are the only questions the POC must answer.

9. Competitive Positioning

9.1 Market Landscape

The AI governance platform market in 2026 falls into four categories, none of which addresses the specific problem the VDA-MD Framework solves:

Category	Key Players	Why They Do Not Compete
Enterprise GRC platforms	Vanta, Drata, Credo AI, OneTrust	Passive monitoring tools. Cannot generate governance rules, govern AI agents, or be operated by non-technical domain owners.
AI gateway vendors	Traefik Hub, Bifrost, Kong AI Gateway, MintMCP	Network-layer traffic control. Govern agent access to tools but not agent behaviour within those tools.
Enterprise AI orchestration	Palantir AIP, IBM Watsonx, Beam AI	Heavy engineering overhead. Require dedicated teams. Not accessible to hotel GMs or clinic owners.
Copilot builders	MS Copilot, Custom GPTs	No governance structure. High hallucination risk. No domain separation, no RACI, no audit trail.

9.2 The VDA-MD Differentiation

The VDA-MD Framework occupies uncontested ground at the intersection of three requirements that no existing product satisfies simultaneously:

- Governance files that a non-technical domain owner can read, edit, and own — not dashboards, not Rego policies, not YAML configurations
- Compliance standards (NIST, ISO, APQC) as the automatic baseline — not a blank canvas that the organisation must fill from scratch
- The customer journey as the organisational accountability architecture — embedding the RACI in the file structure rather than maintaining it separately

The Baseline + Delta model is the specific innovation that creates durable competitive advantage. Every competitor is building ingestion-first platforms that require organisations to document everything. The VDA-MD requires organisations to document only their exceptions — which is both faster to deploy and creates a cleaner, more auditable governance record.

9.3 The Format Advantage and Vendor Independence

By standardising on Markdown as the governance primitive, the VDA-MD Framework ensures that organisations own their governance logic in perpetuity, in a format that is independent of any vendor, platform, or technology stack. The .md files that constitute an organisation's

governance library are plain text. They can be read by any text editor, versioned by any Git provider, consumed by any LLM, and migrated to any future platform without conversion.

This is the opposite of proprietary schema lock-in. The organisation's investment in building a high-quality governance library — encoding its brand voice, operational exceptions, and compliance baselines — does not depreciate when platform decisions change. This is a genuinely durable competitive advantage that strengthens over time as the library grows.

10. Implementation Roadmap

10.1 Phase 0: Foundation

Establish the technical foundation and validate the core toolchain. This phase has no novel risk — all components are mature or validated in the POC.

- Initialize GitHub private repository with VDA directory structure
- Configure Markdown MCP server and validate document conversion pipeline
- Load NIST OSCAL SP 800-53 Rev 5 catalog and ISO 42001 as C2MD source references
- Deploy React dashboard with basic file editing and version history UI
- Configure Claude API access with system prompt establishing VDA context

10.2 Phase 1: C2MD Baseline Generation

Generate the compliance baseline for the target vertical and relevant Shared Services value streams using the C2MD pipeline.

- Run C2MD translation for 8-10 NIST controls relevant to the target vertical
- Generate S2P, O2C, and HRpay baseline files from APQC and NIST sources
- Compliance Officer review and approval of all generated baselines
- Commit approved baselines to GitHub with reviewer metadata in YAML front matter
- Validate that domain owners can read and understand all baseline files without assistance

10.3 Phase 2: Delta Ingestion

Ingest the organisation's unique operational documents using Markdown and generate exception overlays.

- Bulk ingestion of existing SOPs, policy documents, and brand guidelines through Markdown
- LLM classification of each ingested document as baseline-aligned or exception
- Domain owner review and approval of all classified documents
- Exception .md creation for all identified deviations, with full RACI metadata
- First sandbox simulation — validate that exceptions correctly override baselines

10.4 Phase 3: Agent Deployment and Witness

Deploy operational AI agents governed by the VDA-MD library and activate the Witness Agent audit trail.

- Deploy first operational agent in the lowest-risk journey stage (recommend: Post-Stay for hospitality)
- Activate Witness Agent with audit_log.md generation for all agent decisions
- First live exception test — validate that the agent respects cross-domain inheritance rules
- Present first audit log sample to Compliance Officer for evidence quality validation
- Expand to additional journey stages based on Phase 3 learnings

11. Conclusion

The VDA-MD Framework represents a fundamentally different answer to the enterprise AI governance problem — one grounded in thirty years of operational experience with what enterprise technology adoption actually requires, rather than what technology architects assume it requires.

The core insight — assume best practice, govern only the exceptions — sounds simple in retrospect. But it required witnessing the failure of BPR, ERP, and Knowledge Management initiatives that tried to digitise everything to understand why selective digitisation of the genuine delta is the only approach that works at enterprise scale with enterprise budgets.

C2MD is the technical mechanism that makes this insight deployable. By using large language models to translate compliance standards into human-readable, domain-owner-editable Markdown files, C2MD eliminates the specialist skills barrier that has prevented governance from being owned by the people who actually understand the operational domain. The Compliance Officer can read the baseline. The CFO can edit the exception. The audit trail writes itself. No engineers required for routine governance operations.

The customer journey as organisational accountability architecture is the structural innovation that makes this scalable. By encoding the RACI in the file structure — every file with an owner, every cross-domain action with an inheritance rule, every exception with an expiry — the framework makes governance a property of the system rather than a process that runs alongside it.

The POC will validate the two remaining unknowns: C2MD translation quality and LLM exception boundary reliability. Everything else is proven technology assembled in a novel configuration. The architecture will hold. Some implementation details will require refinement. That is what POCs are for.

Organisations that deploy this framework early will not merely achieve AI governance compliance. They will build a structured, auditable, continuously improving record of how their business makes decisions — expressed in a format that every AI system they ever deploy can understand and act upon. That is not a compliance investment. That is an organisational operating system.

NEXT STEP

Execute the six-week POC as designed in Section 8. Validate C2MD translation quality against three NIST controls. Validate exception boundary reliability against five synthetic purchase scenarios. Validate Witness Agent audit trail against Compliance Officer evidence standard. If all three pass: proceed to Phase 0 production deployment. If any fail: the specific failure mode will define the engineering remediation required.

Appendix A: Glossary

Term	Definition
A-VDA-Wrapper	The enterprise configuration combining the VDA-MD governance layer with the A-Wrapper V2 execution engine (AgentGateway, Qdrant, OPA, ArgoCD).
Baseline	The pre-configured, C2MD-generated Markdown representation of a NIST/ISO/APQC best-practice control. Applies to all agents by default.
C2MD	Compliance to Markdown. The LLM-powered translation of machine-readable compliance standards (NIST OSCAL, ISO 42001, APQC) into structured, human-readable, domain-owner-editable Markdown governance files. Original concept, March 2026.
C2P	Compliance to Policy. The existing OSCAL Compass framework (IBM/RedHat) that translates OSCAL into Kubernetes policy engine formats (Kyverno, OCM). Not compatible with the VDA-MD use case.
Cross-domain inheritance	The architectural rule that a vertical agent cannot execute an action crossing into a Shared Services domain without explicitly inheriting permission from the relevant Shared Services Markdown files.
Delta	The organisation-specific exceptions, brand rules, and operational deviations from the baseline. Ingested via MarkItDown and captured as structured Markdown exception files.
Evidence-as-Code	Compliance audit evidence generated automatically as a byproduct of daily operations, not assembled retrospectively.
Exception	An approved deviation from a baseline rule, encoded in an EXCEPTION.md file with owner, approval chain, conditions, and mandatory expiry date.
Journey stage	A named phase of the customer-facing value stream (e.g., Pre-Book, Book, Stay, Post-Stay). Each stage has a designated domain owner and a specific set of governing Markdown files.
MarkItDown	Microsoft's open-source Python utility (MIT license, v0.1.5) for converting PDFs, DOCX, XLSX, and PPTX files into clean Markdown. Deployed as an MCP server in the VDA-MD stack.
MCP	Model Context Protocol. The open protocol developed by Anthropic for connecting AI agents to external tools and data sources. The standard transport layer for the A-VDA-Wrapper.
OSCAL	Open Security Controls Assessment Language. NIST's machine-readable standard for expressing security controls in XML, JSON, and YAML. The primary source for C2MD baseline generation.

Process debt	The accumulation of undocumented, unoptimised, and non-machine-readable operational rules trapped in static legacy documents. The primary barrier to enterprise AI adoption.
RACI	Responsible, Accountable, Consulted, Informed. In the VDA-MD Framework, encoded as YAML front matter in every Markdown file rather than maintained as a separate spreadsheet.
SKILL.md	The Markdown file type defining executable triggers — the specific API calls and system actions an AI agent is permitted to invoke.
SOP.md	The Markdown file type defining the human-readable policy that governs agent behaviour within a specific domain and journey stage.
Value stream	An end-to-end sequence of activities that creates value for a customer or the organisation. The VDA-MD Framework uses three universal horizontal value streams (S2P, O2C, HRpay) and vertical customer journey value streams specific to each industry.
VDA-MD	Value-Driven-AI Markdown Engine. The governance framework described in this document.
Witness Agent	A parallel monitoring agent that observes and logs every operational agent decision, producing an immutable audit_log.md as Evidence-as-Code.

Appendix B: Standards References

Standard	Relevance to VDA-MD
NIST SP 800-53 Rev 5	Primary source for C2MD baseline generation across all operational domains. AC-2, AU-2, SA-4, and related controls form the foundation of S2P, O2C, and HRpay baselines.
NIST OSCAL (CSWP 53)	Machine-readable format providing structured control definitions as C2MD input. NIST's 2025 draft explicitly identifies agentic AI integration as a future direction.
NIST AI RMF	Risk management framework providing the governance structure that the VDA journey map implements organisationally.
ISO/IEC 42001	AI management system standard. VDA-MD baselines include ISO 42001 human oversight requirements for AI systems making consequential decisions.
APQC Process Classification Framework	Industry-neutral process taxonomy providing the foundation for S2P, O2C, and HRpay baseline definitions.
EU AI Act	Regulatory framework requiring demonstrable governance for high-risk AI systems. The VDA-MD Evidence-as-Code approach directly satisfies the documentation and auditability requirements.

GDPR	Data protection regulation with direct relevance to HRpay and O2C domains. PII handling rules are encoded in Shared Services: Legal baseline files.
SOC 2 Type II	The Witness Agent audit_log.md and GitHub version history together provide the continuous evidence trail required for SOC 2 Type II certification.